

Last Time

Dependent Samples

Advantages to dependent sample designs

Effect Sizes for Mean Comparisons

Cohen's d

Hedges' g

Glass' Δ

Power

Factors affecting power

Today

Finishing Effect Sizes

Power

Factors affecting power

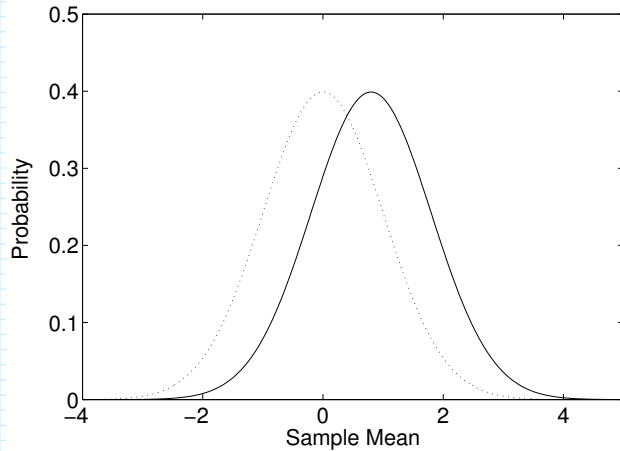
Power Computation

Examples

The Noncentrality Parameter

Determining Sample Size

Effect Size



Cohen's d

Z-test

$$d = \left| \frac{\bar{X} - \mu_0}{\sigma} \right|$$

T-test

$$d = \left| \frac{\bar{X} - \mu_0}{s} \right|$$

Independent Samples T-test

$$d = \left| \frac{\bar{X}_1 - \bar{X}_2}{s_P} \right|$$

Cohen's Convention

d	Size
.2	Small
.5	Medium
.8	Large

GSS Example I

```
> library('effectsize')
> t.test(money,mu=12513)
```

One Sample t-test

```
data: money
t = 13.031, df = 872, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 12513
95 percent confidence interval:
 21548.13 24752.30
sample estimates:
mean of x
 23150.21
```

```
> cohens_d(money,mu=12513)
```

Cohen's d	95% CI
-----------	--------

0.44	[0.37, 0.51]
------	--------------

- Deviation from a difference of 12513.

GSS Example II

```
> t.test(money~happy,alternative="less")

Welch Two Sample t-test

data:  money by happy
t = -3.0283, df = 161.28, p-value = 0.001432
alternative hypothesis: true difference in means between group 0 and group 1 is less than 0
95 percent confidence interval:
    -Inf -2687.825
sample estimates:
mean in group 0 mean in group 1
    17924.87      23849.19

> cohens_d(money~happy,alternative="less",pooled_sd=FALSE)
Cohen's d |          95% CI
-----|-----
-0.28      | [-Inf, -0.12]

- Estimated using un-pooled SD.
- One-sided CIs: lower bound fixed at [-Inf].
```

Power

State of the World	Decision	
	H_0 True	H_0 False
H_0 True	$1 - \alpha$	α
H_0 False	β	$1 - \beta$

Power

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Define significance level:

$$\alpha = Pr(\text{Type I error}) = Pr(\text{Reject } H_0 | H_0 \text{ true})$$

Power

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H_0 False	β	$1 - \beta$

Define significance level:

$$\alpha = Pr(\text{Type I error}) = Pr(\text{Reject } H_0 | H_0 \text{ true})$$

Define power:

$$\begin{aligned} 1 - \beta &= 1 - Pr(\text{Type II error}) \\ &= Pr(\text{Reject } H_0 | H_0 \text{ false}) \end{aligned}$$

Error Probabilities

- ▶ The probability of rejecting a true H_0 is α . We set this probability directly when we select a significance level.
- ▶ The probability of rejecting a false H_0 depends on the effect size! If we specify an effect size, we can compute this probability, which is called *power* and equals $1 - \beta$

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Power is the ability of a test to detect/reject a false null hypothesis.

We can only compute $1 - \beta$ if we know H_1 - if we specify an effect size.

Typically we don't know H_1

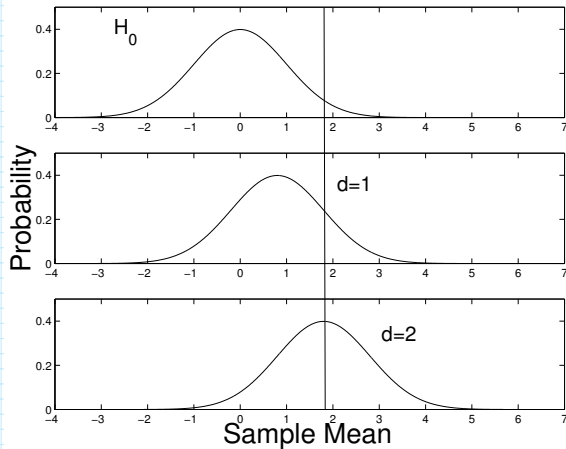
Power and Significance

- ▶ Significance levels α are selected in advance, based on practical considerations.
- ▶ Issues of power are often not considered until after a failure to reject H_0 . However, there are ways to insure good power, which should be used *before* testing H_0 .

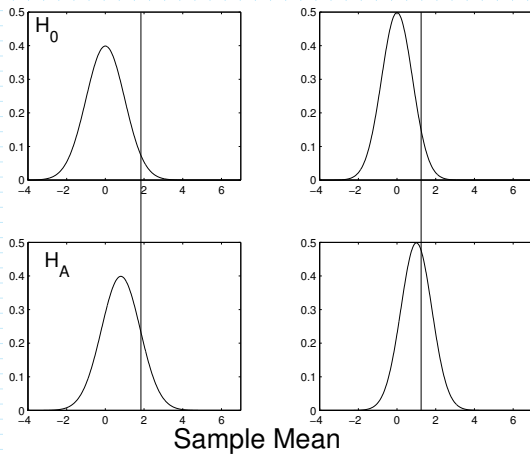
Factors Affecting Power

1. Effect Size (Cohen's d)
2. Sample Size
3. Significance level
 - ▶ One vs. two-tailed tests

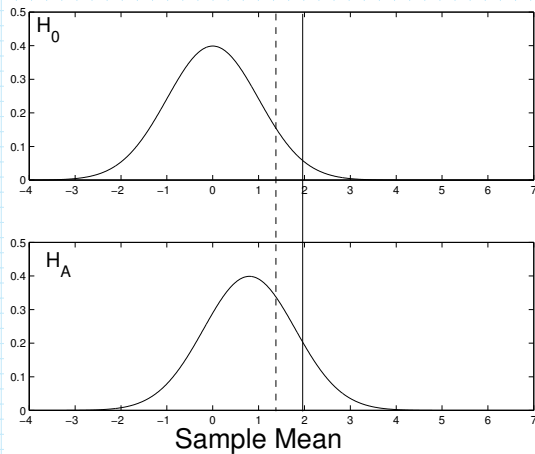
Effect Size



Sample Size



Alpha Level



Power and Standard Error

Another way to increase power: Reduce the variance $\sigma_{\bar{X}_1 - \bar{X}_2}$.

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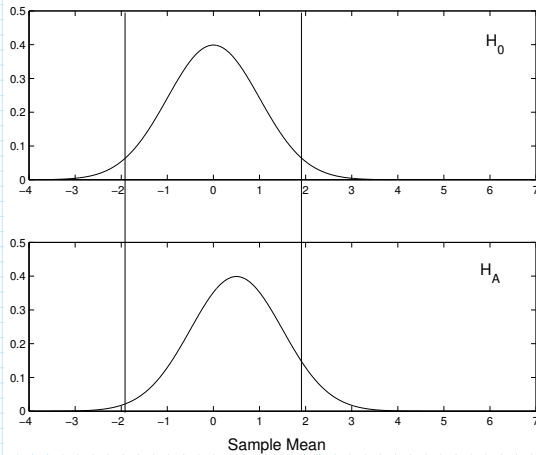
- ▶ Dependent samples

Summary

Power increases with

- ▶ Larger effect sizes
- ▶ Larger sample sizes
- ▶ Larger α -levels (one-tailed tests)
- ▶ Dependent samples designs

A Note on Two-Tailed Tests



Today's Plan

- ▶ Go through z -test power step by step
- ▶ Relate power to effect size for t -tests
- ▶ Demonstrate using R for power computations
- ▶ Determining sample size for desired power and effect size

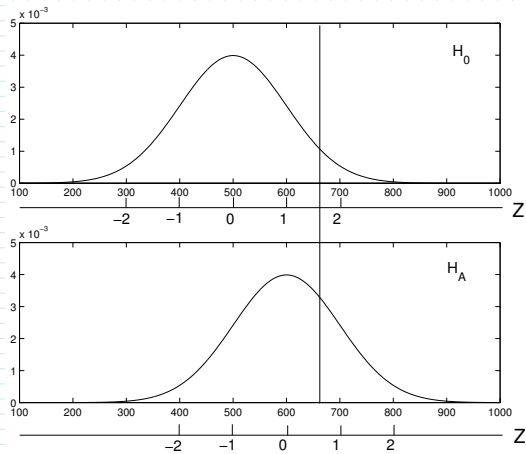
Power Computations

Remember: We need a value for μ_1 (effect size).

$$H_0 : \quad \mu = \mu_0$$

$$H_1 : \quad \mu = \mu_1$$

Setup



Example

In a rat, the number of hippocampal neurons of a certain type has mean $\mu = 500$ and variance $\sigma^2 = 1000^2$. 100 rats are treated with a memory-enhancing drug. On average, their hippocampi contain more of these cells than for untreated rats. If we do a z -test ($\alpha = .05$), what is the power of this test against the alternative that the population mean for treated rats is really 600 cells?

Example

In a rat, the number of hippocampal neurons of a certain type has mean $\mu = 500$ and variance $\sigma^2 = 1000^2$. 100 rats are treated with a memory-enhancing drug. On average, their hippocampi contain more of these cells than for untreated rats. If we do a z -test ($\alpha = .05$), what is the power of this test against the alternative that the population mean for treated rats is really 600 cells?

The desired effect size is $\frac{600 - 500}{1000} = 0.10$.

Problem

► Hypotheses:

$$H_0 : \mu = 500$$

$$H_1 : \mu = 600$$

- What value of $\bar{X}$ would allow us to reject H_0 ?
- What is the z -score of this $\bar{X}$ value *for the alternative distribution*?
- What is the power of the test?

Solution

- ▶ The critical Z value for the test is 1.645
- ▶ The $\bar{X}$ with Z -score 1.645 is

$$1.645 = Z_{\bar{X}} = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}$$

Solution

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- ▶ The Z score of 664.5 *with respect to the alternative hypothesis* is

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- ▶ Power is then

Solution

- ▶ The critical Z value for the test is 1.645
- ▶ The $\bar{X}$ with Z -score 1.645 is

$$1.645 = Z_{\bar{X}} = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}$$

$$\sigma_{\bar{X}} = \sigma / \sqrt{N} = 1000 / \sqrt{100} = 100$$

$$\bar{X} = \mu_0 + (1.645)(100) = 500 + 164.5 = 664.5$$

- ▶ The Z score of 664.5 *with respect to the alternative hypothesis* is

- ▶ Power is then
$$Z_{664.5} = \frac{664.5 - 600}{100} = .645$$

$$Pr(Z > .645) = .2595$$

H_0 is False: The Noncentrality Parameter

$$H_0 : \quad \mu = \mu_0$$

$$H_1 : \quad \mu = \mu_1$$

Consider the test statistic $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{N}}$ when $\bar{X} \sim \mathcal{N}(\mu_1, \sigma^2)$.

Noncentrality Parameter

$$\begin{aligned} E(Z) &= E\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{N}}\right) \\ &= E\left(\frac{\bar{X} - \mu_1 + \mu_1 - \mu_0}{\sigma/\sqrt{N}}\right) \end{aligned}$$

Noncentrality Parameter

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Noncentrality Parameter

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The noncentrality parameter $\delta = d\sqrt{N}$ is the amount by which the test statistic is shifted away from 0 when the null hypothesis is false.

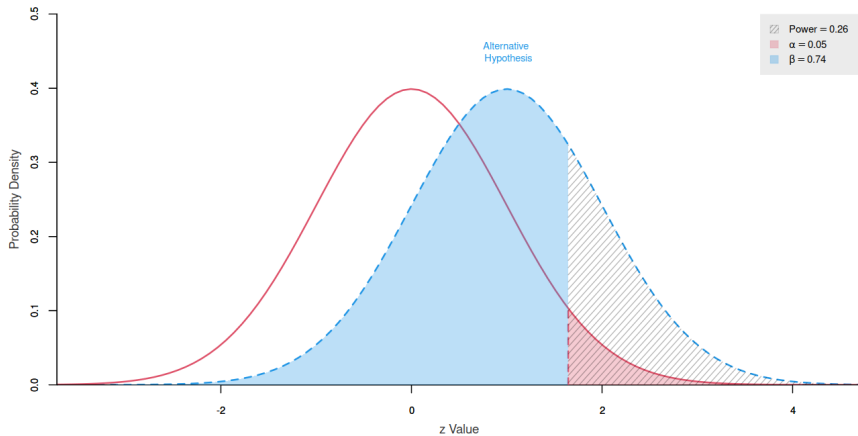
The shape of the t distribution skews when $\delta \neq 0$ and power is no longer easy to compute by hand.

Using R (z-test)

```
> library('pwrss')  
> N <- 100  
> d <- 0.10  
> ncp <- d*sqrt(N)  
> power.z.test(ncp,alpha=0.05,alternative='greater')
```

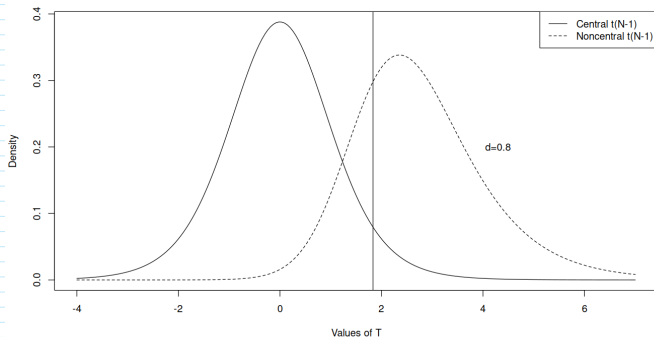
power	ncp.alt	ncp.null	alpha	z.crit
0.259511	1	0	0.05	1.644854

Using R (z -test)



One-Sample t -Test

Problem: the t -distribution shape changes when it is shifted away from 0.
The noncentrality parameter is the same as for the z -test: $d\sqrt{N}$



How well do people answer SAT reading comprehension questions when they haven't read the passage to which the questions refer?

Group 1 Read Passages	Group 2 Don't Read Passages
$\bar{X}_1 = 69.6$	$\bar{X}_2 = 46.6$
$s_1 = 10.6$	$s_2 = 6.8$
$N_1 = 17$	$N_2 = 28$

Effect Size and Noncentrality Parameter

► Effect Size

$$s_P^2 = \frac{(17 - 1)10.6^2 + (28 - 1)6.8^2}{17 + 28 - 2} = 70.84$$

$$s_P = \sqrt{70.84} = 8.42$$

$$d = \left| \frac{69.6 - 46.6}{8.42} \right| = 2.73$$

Effect Size and Noncentrality Parameter

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$$s_P^2 = \frac{(17-1)10.6^2 + (28-1)6.8^2}{17+28-2} = 70.84$$

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$$d = \left| \frac{69.6 - 46.6}{8.42} \right| = 2.73$$

► Noncentrality Parameter

$$\begin{aligned}\delta &= \frac{d}{\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}} \\ &= d\sqrt{\frac{N_1 N_2}{N_1 + N_2}}\end{aligned}$$

Two Independent Samples t -test

We only have the sample statistics and not the individual measurements so we have to do the t -test by hand.

```
> N.1 <- 17
> N.2 <- 28
> mean.1 <- 69.6
> mean.2 <- 46.6

> s.1 <- 10.6
> s.2 <- 6.8
> sp <- sqrt((16*10.6^2 + 27*6.8^2)/(17+28-2))
> df <- N.1 + N.2 - 2

> t <- (mean.1 - mean.2)/(sp*sqrt(1/N.1 + 1/N.2))
> 1 - pt(t,df)

1.36e-11
```

Power in R

```
> d <- abs((mean.1 - mean.2)/sp)
```

```
> d
```

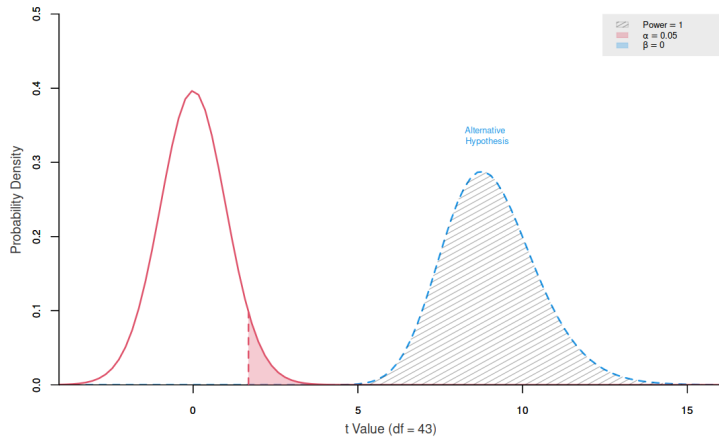
```
[1] 2.732625
```

```
> ncp <- d * sqrt((N.1*N.2)/(N.1 + N.2))
```

```
> power.t.test(ncp,df=df,alpha=.05,alternative='greater')
```

power	ncp.alt	ncp.null	alpha	df	t.crit
1	8.887	0	0.05	43	1.681071

Power in R



Another Example

Scores on the SAT subtests are normally distributed with mean $\mu = 500$ and standard deviation $\sigma = 100$. Sylvan Learning Systems claimed that their study techniques resulted in a mean increase of 100 points on math subtest. If this were true, how many Sylvan students would you have to test to reject $H_0 : \mu = 500$ at an $\alpha = 0.01$ level 95% of the time?

Problem

► Hypotheses:

$$H_0 : \mu = 500$$

$$H_1 : \mu = 600$$

Problem

- Hypotheses:

$$H_0 : \mu = 500$$

$$H_1 : \mu = 600$$

- The critical Z value for the test is 2.33
- The $\bar{X}$ with Z -score 2.33 satisfies

$$2.33 = Z_{\bar{X}} = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}$$

Sample Size Problem

- ▶ The Z score of this $\bar{X}$ *with respect to the alternative hypothesis* is

$$Z_{\bar{X}} = -1.645 = \frac{\bar{X} - 600}{100/\sqrt{N}}$$

Sample Size Problem

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$$Z_{\bar{X}} = -1.645 = \frac{\bar{X} - 600}{100/\sqrt{N}}$$

- ▶ Solve for N :

$$500 + 2.33(100/\sqrt{N}) = 600 - 1.645(100/\sqrt{N})$$

Sample Size Problem

- ▶ The Z score of this $\bar{X}$ *with respect to the alternative hypothesis* is

$$\begin{aligned}Z_{\bar{X}} = -1.645 &= \frac{\bar{X} - 600}{100/\sqrt{N}} \\ \Rightarrow \bar{X} &= 600 - 1.645(100/\sqrt{N})\end{aligned}$$

- ▶ Solve for N :

$$\begin{aligned}500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500\end{aligned}$$

Sample Size Problem

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- Solve for N :

$$\begin{aligned}500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \\ 397.5/\sqrt{N} &= 100\end{aligned}$$

Sample Size Problem

- The Z score of this $\bar{X}$ *with respect to the alternative hypothesis* is

$$\begin{aligned}Z_{\bar{X}} = -1.645 &= \frac{\bar{X} - 600}{100/\sqrt{N}} \\ \Rightarrow \bar{X} &= 600 - 1.645(100/\sqrt{N})\end{aligned}$$

- Solve for N :

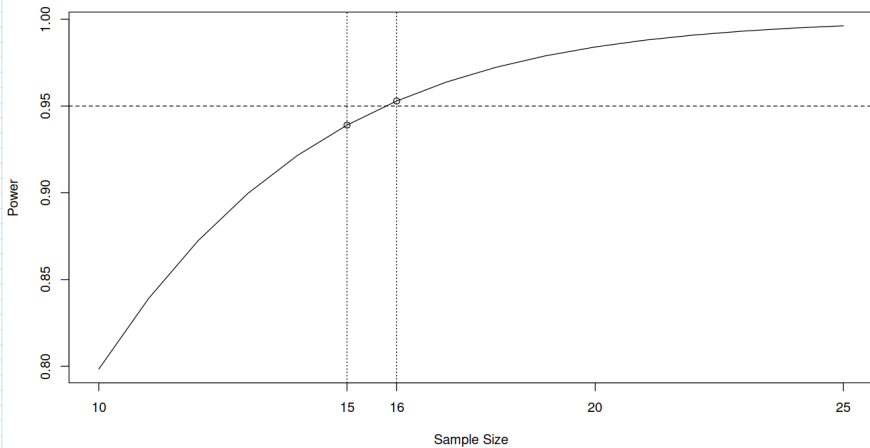
$$\begin{aligned}500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \\ 397.5/\sqrt{N} &= 100 \\ \sqrt{N} &= 3.975 \\ N &= 3.975^2 = 15.80\end{aligned}$$

Sample Size Determination In R

```
> library('pwrss')
> desired.power <- 0.95
> d <- 1.0
> N <- seq(10,25)
> ncp <- d * sqrt(N)
> power <- power.z.test(ncp,alpha=alpha,alternative='greater',plot=FALSE)

> first<-which(power>=desired.power)[1]
> plot(N,power,xlab="Sample Size",ylab="Power",type='l')
> points(N[(first-1):first],power[(first-1):first])
> abline(h=desired.power,lty=2)
> abline(v=c(N[first],N[first-1]),lty=3)
> axis(1,at=16,tick=TRUE)
```

Power Curve



Sample Size for z -Test

```
> alpha <- 0.01
> desired.power <- 0.95
> d <- 1.0 # effect size to be detected
> pwrss.z.mean(mu=d,alpha=alpha,alternative='greater',power=desired.power)
```

A Mean against a Constant (z Test)

H0: $\mu = \mu_0$

HA: $\mu > \mu_0$

Statistical power = 0.95

n = 16

Alternative = 'greater'

Non-centrality parameter = 3.971

Type I error rate = 0.01

Type II error rate = 0.05

Two-sample t -Test

```
> alpha <- 0.05
> desired.power <- 0.80
> d <- 0.5

> power <- pwrss.t.2means(d,alpha=alpha,alternative='greater',
                           kappa=1,power=desired.power,welch.df=TRUE)
```

Difference between Two means
(Independent Samples t Test)

H0: $\mu_1 = \mu_2$

HA: $\mu_1 > \mu_2$

Statistical power = 0.8

n1 = 51

n2 = 51

Alternative = 'greater'

Degrees of freedom = 100

Non-centrality parameter = 2.525

Type I error rate = 0.05

Type II error rate = 0.2

Power Curves and Tables

Student's *t*-Distribution

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NUMBER OF OBSERVATIONS FOR *t*-TEST OF DIFFERENCE BETWEEN TWO MEANS

Single-sided test Double-sided test	Level of <i>t</i> -test																			
	$\alpha = 0.005$					$\alpha = 0.01$					$\alpha = 0.025$					$\alpha = 0.05$				
	$\alpha = 0.01$					$\alpha = 0.02$					$\alpha = 0.05$					$\alpha = 0.1$				
$\beta =$	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5
0.05																				0.05
0.10																				0.10
0.15																				0.15
0.20																			137	0.20
0.25															124				88	0.25
0.30										123					87				61	0.30
0.35										90					64			102	45	0.35
0.40										70					50			108	78	0.40
0.45										55				105	79	39		108	86	0.45
0.50										45				106	86	64	32	88	70	0.50
0.55										38				87	71	53	27	112	73	0.55
0.60										32				74	60	45	23	89	61	0.60
0.65										27				63	51	39	20	76	52	0.65
0.70										24				55	44	34	17	66	45	0.70
0.75										21				48	39	29	15	57	40	0.75
0.80										19				42	34	26	14	50	35	0.80
0.85										17				37	31	23	12	45	31	0.85
0.90										15				34	27	21	11	40	28	0.90
0.95										14				30	25	19	10	36	25	0.95
1.00										13				27	23	17	9	33	23	1.00
1.1										11				23	19	14	8	27	19	1.1
1.2										9				20	16	12	7	23	16	1.2
1.3										8				17	14	11	6	20	14	1.3
1.4										8				15	12	10	6	17	12	1.4
1.5										7				13	11	9	5	15	11	1.5
1.6										6				12	10	8	5	14	10	1.6
1.7										6				11	9	7	4	12	9	1.7
1.8										5				10	8	6	4	11	8	1.8
1.9										5				9	7	5	3	10	7	1.9
2.0										4				8	6	4	2	9	6	2.0
2.1										4				7	5	3	1	8	5	2.1
2.2										4				6	4	2	1	7	4	2.2
2.3										3				5	3	1	1	6	3	2.3
2.4										3				4	2	1	1	5	2	2.4
2.5										3				4	2	1	1	5	2	2.5
2.6										3				3	1	1	1	4	1	2.6
2.7										3				3	1	1	1	4	1	2.7
2.8										3				3	1	1	1	4	1	2.8
2.9										3				3	1	1	1	4	1	2.9
3.0										3				3	1	1	1	4	1	3.0
3.1										3				3	1	1	1	4	1	3.1
3.2										3				3	1	1	1	4	1	3.2
3.3										3				3	1	1	1	4	1	3.3
3.4										3				3	1	1	1	4	1	3.4
3.5										3				3	1	1	1	4	1	3.5
3.6										3				3	1	1	1	4	1	3.6
3.7										3				3	1	1	1	4	1	3.7
3.8										3				3	1	1	1	4	1	3.8
3.9										3				3	1	1	1	4	1	3.9
4.0										3				3	1	1	1	4	1	4.0
4.1										3				3	1	1	1	4	1	4.1
4.2										3				3	1	1	1	4	1	4.2
4.3										3				3	1	1	1	4	1	4.3
4.4										3				3	1	1	1	4	1	4.4
4.5										3				3	1	1	1	4	1	4.5
4.6										3				3	1	1	1	4	1	4.6
4.7										3				3	1	1	1	4	1	4.7
4.8										3				3	1	1	1	4	1	4.8
4.9										3				3	1	1	1	4	1	4.9
5.0										3				3	1	1	1	4	1	5.0

Other Resources

- ▶ R pwr package
- ▶ R stats package (`power.t.test`, `power.prop.test`, `power.anova.test`)
 - ▶ Note that `power.t.test` in the `pwrss` package overwrites the `power.t.test` function in the stats package that is loaded into R by default.
- ▶ G*Power (Mac and Windows)